

Learning outcomes & mapping

The ten learning outcomes for this session, each mapped to the slide(s) that teach it, the notebook section where the trainee *does* it, and how it is **assessed** (how you — and the trainee — can tell it landed).

By the end of the session, trainees should be able to:

#	Learning outcome	Slide(s)	Notebook section	How it's assessed
1	Explain why RAG is needed (parametric knowledge, staleness, private data)	3, 4, 5	Intro + Section 2 (why a knowledge base at all)	Misconception 1; cold-call “why can’t the base model answer about <i>your</i> runbook?”; recap slide 21
2	Describe the RAG architecture and name each stage	6, 7, 8, 9	Whole notebook; the “toy → production” bridge table	Trainee redraws/labels the Slide 8 pipeline from memory; applies the Slide 9 “inspect retrieval first” rule
3	Explain document chunking (size, overlap, boundaries)	11, 12	Section 4 (chunking + the overlap demo cell)	Stretch 2 (chunk-size U-curve); Misconception 2; trainee points to overlapping words across a boundary
4	Explain embeddings and vector similarity	13, 14	Section 5 (embeddings), Section 6 (cosine)	Stretch 4 (embeddings bake-off); Misconception 3; trainee reads the score gap between relevant and irrelevant chunks
5	Implement basic Top-K retrieval	15	Section 7 (retrieve(query, top_k))	Trainee runs/extends retrieve; Stretch 1 (Top-K); explains argsort → Top-K
6	Construct retrieved context for an LLM	16	Section 8 (build_context, assembled prompt)	Trainee inspects the printed prompt and explains why instructions / evidence / question are kept separate
7	Generate a grounded answer	17	Section 9 (generate_answer), Section 10	Runs the grounded answerer; the Lumen CLI fictional-tool demo proves the answer came from retrieval, not memory
8	Explain why citations matter	18	Section 10 (format_with_citations)	Misconception 5; trainee states “provenance, not proof” and what a citation does <i>not</i> guarantee
9	Identify common RAG failure modes	9, 19	Section 11 (Mars refusal)	Stretch 3 (grounding guard); Misconceptions 2–4; trainee triggers and explains a refusal

#	Learning outcome	Slide(s)	Notebook section	How it's assessed
10	Explain why RAG does not guarantee factual correctness	18, 19	Section 11 + bridge (conflicting-evidence note)	Misconceptions 4 & 5; discussion: "if the retrieved source is itself wrong, what does a perfect RAG system do?"

How assessment works in this session

This is a 90-minute hands-on session, so assessment is **formative and observational**, not a graded test. Four lightweight instruments, already built into the pack, give you evidence for every outcome above:

The **notebook outputs** are the primary evidence for the "doing" outcomes (3–7, 9): if a trainee's cells run and produce the expected retrieval scores, grounded answer, citations, and refusal, they have demonstrated the mechanics end to end.

The **stretch challenges** ([stretch-tasks.pdf](#)) are the deeper probes. Stretch 1 assesses Top-K (outcome 5/9), Stretch 2 assesses chunking (outcome 3), Stretch 3 assesses failure handling and refusal (outcomes 9, 10), and Stretch 4 assesses embeddings (outcome 4). The **answer key** tells you what mastery looks like and which wrong conclusions to correct.

The **five misconceptions** ([misconceptions.pdf](#)) are your discussion-based checks for the "explain" outcomes (1, 2, 8, 10). Posing a misconception and asking the room to correct it reveals understanding faster than a direct question.

The **refusal demo** (notebook Section 11, Slide 19) is the single best check for the two hardest outcomes (9 and 10): a trainee who can explain *why* the Mars question must be refused — and what that implies about trust — has understood the heart of the session.

Outcomes 1, 2, 8, and 10 are *conceptual* and are assessed by explanation/discussion. Outcomes 3–7 and 9 are *practical* and are assessed by the notebook and stretch challenges. Every outcome has at least one slide **and** one notebook touchpoint, so no outcome is taught only in the abstract.